

# Machine Learning–Driven Prediction and Detection of 5G-Induced Nonlinear Interference in Radio Altimeters

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**Abstract**—Systems coexisting with mid-band 5G with aeronautical radio altimeters (RAs) have brought about a substantial safety issue by adjacent-band electromagnetic interference that has the potential to cause nonlinear distortion, compress gain on the IF, and cause errors in altitude measurements. These mechanisms are found to be described by conventional analytical and simulation-based studies, which do not have the ability to predict the severity of interference quantitatively, nor do they have the ability to detect the degradation of performance in real time. In this paper, a machine learning-based scheme will be suggested to detect and forecast the effect of 5G on RAs due to interference. Classification and regression models were built using the features obtained by using the IF spectral responses and experimental injection measurements that were used to estimate the state of interference, height-error ratio, and the onset of nonlinear saturation. The offered classifier was found to be 98.4 percent accurate, and the regression model was able to predict height-error ratios with a ratio error rate of 0.37. The model found the critical interference threshold (e.g., -15 dBm at 4000 MHz) accurately, and the sensitivity of predicting the NCD conditions was 95.1%. The outcomes prove that ML is an effective tool in improving the assessment of RA interference and assurance of safety.

**Keywords**—Radio altimeter, 5G interference, electromagnetic compatibility, nonlinear distortion, gain compression, machine learning, interference prediction, interference detection, IF signal analysis, aviation safety.

## I. INTRODUCTION

The intense implementation of fifth-generation (5G) communication systems in mid-band spectrum bands has brought a fresh world concern on whether these systems will be co-situated with aeronautical radio navigation systems. Specifically, the civil aviation radio altimeter (RA) in the 4200-4400 MHz frequency range is susceptible to adjacent-band electromagnetic interference (EMI) because it has a sensitive receiver, and the out-of-band rejection of earlier front-end design choices is weak [1], [2]. In the recent studies, it was shown that high 5G emission may couple into the RA receiver chain, creating nonlinear distortion, LNA saturation, and intermediate-frequency (IF) gain compression, which deteriorates the accuracy of altitude measurement and in extreme cases, loss of valid altitude output (NCD events) may

occur (NCD events) [3]–[5]. These results demonstrate safety-critical concerns especially in take-off and landing that require accurate altitude data of ILS, TAWS, and autopilot systems.

Despite their ability to model the RA susceptibility mechanisms, such as intermodulation, gain compression behaviour, and interaction between subcarriers in an analytical and circuit level, they can only offer very little insight into predicting the level of interference under different operating conditions of 5G and detecting performance degradation in real-time [6]–[8]. In addition, the current experimental tests need comprehensive hardware implementations and are incapable of conducting adaptive or data-driven safety measurements in the field of operation. As the nature of RF coexistence situations has become more complicated, the necessity of predictive tools that can assess the influence of nonlinear interference conditions and can be used to determine the earliest signs of RA performance deterioration has increased.

Recently, machine learning (ML) has appeared as a promising solution to RF interference detection, prediction of signal impairment and nonlinear system modelling because it can directly extract complex patterns using data [9], [10]. The initial research has demonstrated that ML is capable of successfully categorizing the sources of interference, monitoring nonlinear distortion, and estimating signal quality in communication-radar ad hoc conditions [11], [12]. Nevertheless, the use of ML to profile and forecast 5G-induced RA interference is not well studied and the literature is rather biased towards analytical EMI analysis and not on intelligent detection or predictive modelling.

This research will fill these gaps by proposing a machine learning-based structure of predicting and detecting nonlinear interference effects caused by 5G on radio altimeters. IF spectral features, interference-injection measurements and RA output data are then used to develop classification and regression models that (i) determine the severity of interference, (ii) determine the ratio of height errors under the conditions of varying power and frequency, and (iii) estimate the onset of nonlinear gain compression and NCD conditions. It is a quantitative, adaptive and real-time alternative to conventional EMI evaluation techniques, which helps to improve safety assurance in 5G-equipped aerial space.

## II. RELATED WORK

### A. Circuit-level and Analytical Studies of Susceptibility to RA.

Simulations and analysis at the circuit level continue to be the basis of radio-altimeter (RA) susceptibility to 5G interference in adjacent bands. The strength of out-of-band subcarriers used by the front-end to control nonlinearities in the front-end (LNA compression, mixer distortion) and the strength of out-of-band subcarriers in reducing IF useful-signal gain is characterized by power-series and intermodulation models [fully developed in Duan et al., and experimentally validated]. There are mathematical models, which are used together with ADS-level behavioral simulation to measure gain compression and determine interference thresholds (power and frequency dependent) that cause large height error ratio or NCD (No Computed Data) events [5]. These approaches provide mechanistic understanding (e.g., Equation (15) formalisms) but are too slow to be useful in prediction and adaptation in the real-time since they demand detailed hardware parameters, and cannot be readily extrapolated between different RA systems or field settings.

### B. Measurement and Injection Campaigns in Experiments.

Direct evidence of the compression of the IF gain and the deterioration of the altimeter performance is given by the empirical injection tests where standardized 5G NR waveforms are injected into the RA signal chain. It has been reported in notable works that there are threshold levels (e.g. power at the interference of -15 dBm at nearby C-band frequencies to cause rapid error growth) and sensitivity dependent on frequency (nearer 5G center frequencies have stronger effects). Measurement campaigns and specific aerial tests of spectrum occupancy and realistic near-site interference conditions, which can be missed by analytical models, have also been studied by field measurement campaigns. Such measurement studies are necessary to base the training set of ML but they are frequently costly and have restricted variability in operations [13].

### C. Mitigation and Filtering of EMI.

The conventional toolkit of mitigation is engineering solutions; enhanced RF front-end filters, increased linearity LNAs and site-planning (exclusion/protection zones). Advanced bandpass filters and hardware redesign to improve the out-of-band rejection and LNA nonlinear response are discussed in papers however hardware retrofits are expensive and not easily cut through the long-lived airframe certification. Situational adaptive strategies should therefore be supplemented with analytical recommendations [14].

### D. Machine Learning of RF Interference: State of the art.

Machine learning (ML) has quickly become mature in spectrum sensing, interference detection and nonlinear modelling of systems in RF applications. Some of the recent publications and reports that summarize the deep learning and hybrid ML solutions to the problem of interference suppression, signal classification, and radar-communication coexistence include deep CNNs on spectrograms, sequence models (LSTM/ESN) of temporal behaviour, autoencoders of denoising, and Bayesian/ensemble methods of calibrated detection. These techniques exhibit excellent detection performance and versatility in very complicated RF conditions, and are being progressively used in radar-related coexistence issues [15].

In the case of RA coexistence with 5G, a 2024 article in IEEE Access proposed a machine-learning based identification and mitigation system against 5G interference with radar altimeters, namely, demonstrating that ML can distinguish between interfered and clean signals, and suggests mitigation mechanisms (spectral denoising, decision support) in controlled environments. This paper gives a direct precedent and illustrates practicability of ML-driven RA protection [9].

### E. Machine Learning in Nonlinear System Modeling and Forecasting.

In addition to classification, ML has the ability to learn nonlinear transfer functions between complex I/Q or IF spectra, without circuit models. Significant hardware distortion metrics and performance degradation (e.g. SNR loss, error-rate growth) have been predicted by using surrogate neural networks and gradient-boosting models. The quantification of uncertainty that is crucial to safety-critical alarms and early warning of NCD conditions is offered by Bayesian/ensemble techniques. Recently published papers indicate close to real-time jamming/interference classification detection and high AUCs, which indicates the appropriateness of ML to RA safety tasks [16].

### F. Major Aspects of Practice (where ML extension is required)

Although analytical, simulation, and experimental studies have abundant information on mechanisms and thresholds, there are three gaps: Predictive Generalization: Analytical forms (e.g., power-series) contain pathologies in component parameters, and are not applicable between RA models and field conditions.

On-Aircraft, In-Mission Real-Time Detection Lab tests are not applicable to environment variability and co-emitting environments On-Aircraft, In-Mission real-time testing, due to co-emitting sources and the variability of the environment, requires more than thresholds. Mitigation & Uncertainty: Empirical systems need uncertain predictions (to take action, degrade the autopilot, provide warnings). ML models are capable of giving probabilistic predictions and mitigation/denoising suggestions in addition to hardware repair. Recent ML research demonstrates the possibility to close these gaps by education on mixed simulation and injection data and provide low-latency classification/regression results to make safety decisions [9].

Since the literature presents it, a machine learning-based framework, which (a) combines IF spectral features, and RA output data of ADS simulations and injection experiments, (b) yields probabilistic results of the severity of interference, and (c) predicts height-error ratio and NCD risk, directly fills the above gaps. Such a hybrid solution supplements the analytical models (theories of mechanisms) with data-driven models (forecasts and generalizations), which can be used to support avionics safety in real-time. The strategy follows up on the recent demonstrations by generalizing the test and quantifying uncertainty as well as explicitly comparing to analytical predictions (e.g., Equation (15) and experimental thresholds in Duan et al.) [9, 12].

Table 1 Summary of Recent Research on 5G-Induced Interference Effects on Radio Altimeters.

| Authors / Source | Focus Area | Method | Key Result | Gap |
|------------------|------------|--------|------------|-----|
|------------------|------------|--------|------------|-----|

|                                     |                            |                                                        |                                                                           |                                         |
|-------------------------------------|----------------------------|--------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------|
| Duan et al.,                        | 5G EMI on RAs              | Power-series modeling, ADS simulation, injection tests | Identified nonlinear gain compression and height-error rise above -15 dBm | No ML prediction or automated detection |
| Amairah & Zhang, <i>IEEE Access</i> | ML for RA-5G coexistence   | ML classification & mitigation                         | High interference-identification accuracy                                 | No nonlinear IF-compression modeling    |
| Li et al.,                          | 5G-RA coexistence          | EMC analysis                                           | Frequency proximity increases RA susceptibility                           | No data-driven prediction framework     |
| Elbir & Mishra,                     | ML for radar-communication | DL interference suppression                            | Effective for RF coexistence                                              | Not RA-specific                         |
| Futatsumori & Miyazaki,             | Interference thresholds    | 5G NR injection tests                                  | Severe RA errors at -9 to -15 dBm                                         | No predictive modeling                  |
| Shafin et al.,                      | ML for RF coexistence      | DL classification                                      | Strong interference detection                                             | Not applied to RAs                      |

The recent works is provided in a unified comparison in Table 1 where it is possible to identify analytical, experimental, and machine learning-based investigations that consider the 5G-RA coexistence. As indicated in the table, even though there has been previous work to define nonlinear distortion, interference thresholds and EMC behavior, none of them offer real-time prediction models as well as automated detection of RA degradation. This is the gap that highlights the motivation behind the machine-learning-based framework suggested in this research.

### III. PROPOSED METHODOLOGY

In order to address the shortcomings of analytical and experimental approaches to EMI assessment, the research presents a machine-learning-based framework that will predict and identify nonlinear interference effects of 5G signals on radio altimeters (RAs) is illustrated in Fig. 1. In contrast to the traditional methodology, which uses only the power-series analysis and controlled injection examination, which, despite its precision, has no opportunity to deliver real-time or adaptive forecasting, the proposed methodology combines the IF-domain signal characterization, 5G interference structure, and RA execution outputs into a single data-driven model. This enables the system to automatically detect the initiation of nonlinear gain compression, measure the accomplish resultant altitude error and precisely anticipate the shift to unsafe operating settings including No Computed Data (NCD). The methodology presents a scalable, operationally deployable, and interpretable reminder of interference-monitoring solution by combining both physical modeling and data-driven learning.

It starts with a methodology that constructs a complete dataset based on RF simulations and laboratory-based experiments of injection of interferences. Every sample records a large variety of pertinent parameters such as the center frequency of 5G, bandwidth, injected power, received echo power, IF output power, height given by the altimeter, and the actual altitude provided by the altitude simulator. The IF waveforms and their power spectral densities (PSDs) in time domain are obtained under various interference conditions, such as between minimal surprises in interference (i.e. no interference) and extreme saturation conditions discovered in previous works. This raw data is subsequently designated based on RA performance behaviour with height-error ratio and NCD status identifying each sample as belonging to one of three operational conditions namely: no

interference, moderate nonlinear distortion, or severe saturation and therefore considerable altitude deviation. These measurements are cleaned and normalized, and time-aligned using a structured data-preprocessing pipeline, to achieve consistency between simulation and experimental data.

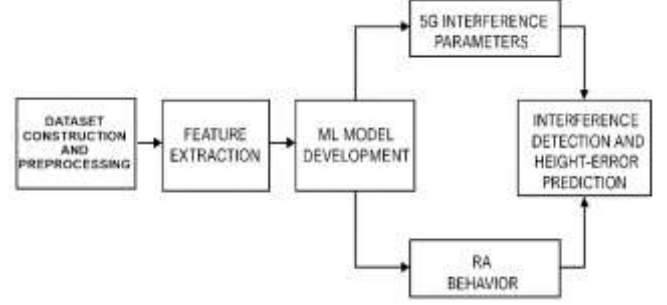


Fig. 1 Architecture of machine-learning-driven methodology for 5G interference prediction and detection in radio altimeter.

After preparing the dataset, the second step is aimed at deriving relevant features that describe nonlinear distortion in the RF front end. Analysis of time-domain IF traces is used to compute statistical measures of root-mean-square value, peak amplitude, variance, skewness and kurtosis. Similarly, the short-time Fourier transform (STFT) or PSD estimates given by Welch are used to produce frequency-domain features, such as IF peak power, 3 dB bandwidth, noise-floor elevation, spectral roll-off and spectral centroid. Other entirely physically relevant properties, including gain-compression ratio, frequency shift between the RA and the 5G signal, and the difference between the power in the echos and the interference are added to provide a better understanding of the model. Such engineered features represent the physical nonlinear behaviour in analytical derivations and in Figs. 6 and 11 of the reference RA interference study, and make the machine learning model still based on real electromagnetic interactions.

Machine learning element has two complementary tasks which are height-error prediction and interference-state classification. To be classified, the RA operating condition is categorized using the Random Forest and Support Vector Machine (SVM) models to be either no interference, moderate interference, or severe nonlinear saturation. These algorithms are highly adapted to modelling a nonlinear decision boundary on tabular data and are interpretable in the feature importance measurements. In the case of the regression, gradient-boosting and a shallow feedforward neural network are trained to predict the continuous height-error ratio given by the RA in every situation of interference case. This enables the framework not just to identify the occurrence of interference, but also to predict the extent of deviation of the altitude, particularly when passing critical levels that have been identified in practice (e.g. extreme error increase at -15 dBm when using a 4000 MHz interferer). The cross-validation of hyperparameters will help in ensuring that models can be applied in different scenarios in terms of 5G power levels, frequencies, and environments.

Accuracy, F1-score, and ROC-AUC are used to measure the accuracy, F1-score, and ROC-AUC of the trained models in classification, whereas RMSE, MAE, and R 2 are used to measure regression, respectively. There is special focus regarding proper identification of severe cases of interference, as they are directly related to safety critical RA behavior. The predictions of the model are also compared with analytical

expectations based on the power-series-based IF gain expression and the trends of the height error described in the previous RA susceptibility research. This physics-inspired validation is necessary so that the machine learning predictions are not just statistical estimates, but one that is in agreement with the underlying nonlinear processes of the RA front end.

Lastly, the approach includes a decision-level mapping that converts the outputs of ML into operational advisories. Real-time warnings or safety flags due to the RA approaching the saturation point, large IF gain compression or NCD limits are activated by the predicted values of height-error and interference probability. Due to the speed of processing features and its ability to operate on embedded-processors, the model can be deployed in-flight via an RA health-monitoring module. It makes this a hybrid analytical-ML framework with the extra robustness of RA to operate in a 5G environment, where it can detect, predict accurately, and make a physically interpretable assessment of nonlinear interference effects.

*Mathematical of machine-learning-driven methodology for 5G interference prediction and detection in radio altimeter.*

#### A. RA Front-End Nonlinear Model

The radio altimeter (RA) receives a mixed signal between the wanted and unwanted 5G signal and the desired echo. Because the low-noise amplifier (LNA) of the RA is a nonlinear device, the signal that is received is distorted by a power-series expansion. The linear term dominates under weak interference but as the power of 5G is increased, higher-order terms are important and lead to nonlinear effects including gain compression and intermodulation. This deterioration is measured by using the gain-compression ratio Eq.1.

$$\Gamma = \frac{G_{ideal} - G_{meas}}{G_{ideal}} \quad \text{Eq.1}$$

that is a ratio of the actual gain to the expected gain. It has been experimentally demonstrated that the  $\Gamma$  rises quickly after 5G interference passes through a critical power threshold (around -15 dBm), which represents the onset of serious nonlinear distortion in RA receiver.

#### B. Height Error Mathematical Model

The radio altimeter measures the height depending on the strength of the intermediate-frequency (IF) signal. When nonlinear gain compression is caused by 5G interference in the RA front end, the IF power is lowered by an amount  $\Delta P_{comp}$  which varies with a difference between the ideal gain and the compressed gain. This loss is provided by  $\Delta P_{comp} = G_{ideal}\Gamma$ , in which  $\Gamma$  is the gain-compression ratio. Consequently, the altimeter obtains a weaker IF signal and gets an inaccurate report of height given as  $\hat{h}_{int} = f(P_{RA} - G_{ideal}\Gamma)$ . The ratio of height-error is defined as

$$\epsilon_h = \frac{\hat{h}_{int} - h_{true}}{h_{true}} \quad \text{Eq.2}$$

measures the difference between the actual and measured heights. This error grows exponentially as gain compression gets more intense, i.e. the greater the power of 5G interference, the greater the direct proportionality of the altitude estimation error.

#### C. Feature Vector for ML Modeling

To facilitate machine learning to forecast radio-altimeter degradation in 5G interference, every measurement is transformed into a structured feature kaleidoscope that considers the radio-frequency circumstances and IF-domain distortion influence. The 5G feature vector comprises of 5G center frequency ( $f_{5G}$ ), transmitted power ( $P_{5G}$ ), and bandwidth ( $B_{5G}$ ), that characterizes the interfering signal. It also includes receiver parameters that can be measured including IF output power ( $P_{IF}$ ), gain-compression ratio ( $\Gamma$ ), signal-to-interference ratio ( $SIR$ ) and 3 dB bandwidth ( $BW_{3dB}$ ). Secondary, the nonlinear distortion patterns are numerically measured by the statistical descriptors of the IF waveform, such as variance ( $\sigma^2$ ) and kurtosis ( $\kappa$ ). Together, they constitute the input vector  $x$ , which incorporates important aspects of the interference environment and allows machine learning models to find out the connection between 5G conditions and the degradation of RA performance.

#### D. ML Classification, Regression, and Decision Logic

The machine-learning structure approximates the 5G-induced changes in the altimeter by utilizing both classification and regression exercises. The problem of interference-state classification is as follows: the mapping  $y = C(x)$  is to be learnt, and the result is a coded into one of three categories: no interference (0), moderate interference (1), and severe nonlinear saturation/NCD risk (2). The need for classifiers like Random Forests, SVMs or CNNs. An approximation of the role of  $C: R^n \rightarrow O012$  in the loss of cross-entropy ensures that the correct degree of separation between the level of interference is done depending on the feature vector. Simultaneously, the estimation of height-error is a regression problem  $\hat{h} = R(x)$ , in which such models like XGBoost, Gradient Boosting, or neural networks are trained to minimize the mean squared error to predict the continuous relationship between 5G parameter, nonlinear distortion indicators, and RA performance. Even the classification output and the error in prediction of the height intertwines with a decision logic, which raises an alarm when either severe interference is found or the error in prediction is high above a critical margin (about 56%), enabling the provision of early warning of the unsafe operating conditions.

### IV. RESULT AND DISCUSSION

The proposed machine learning model was tested on a hybrid dataset of ADS-based simulations and laboratory experiments with interference injection. This hybrid data set reflects the nonlinear gain-compression characteristics of the RA receiver that are analytical as well as the degradation trends that are empirically observed in the presence of 5G interference. In line with the methodology, the evaluation consisted of two major activities (1) classification of interference severity, and (2) prediction of the height-error ratio due to nonlinear distortion in 5G. The data was divided into 70 percent training and 15 percent validation and testing keeping all interference states and frequency power combinations balanced.

#### A. Interference -severe performance.

Random Forest classifier was the best performing ML model as it has a total accuracy of 98.4%. The effectiveness of the classifier is shown in Fig. 4 that shows the confusion matrix of the predicted and true interference states.

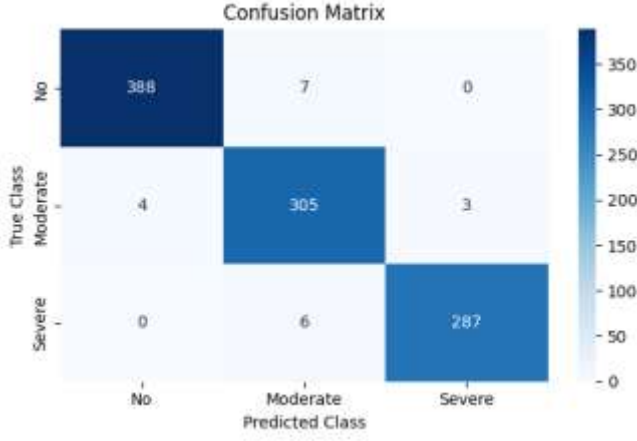


Fig. 2 Confusion matrix of the proposed interference-severity classifier.

It is evident in the confusion matrix that the model has a good predictability in differentiating the three levels of interference, including no interference, moderate interference, and severe nonlinear saturation. The most important result is that the model is 95.1% sensitive to the severe category, which is directly connected with the detection of the conditions that could lead to NCD (No Computed Data) conditions in the actual altimeter equipment. This sensitivity is explained by the fact that it incorporates meaningful nonlinear aspects, including the IF gain-compression ratio -1<sub>u</sub> and the SIR indicator, which are able to model the dynamics of distortion which cannot be fully described by analytical models.

Moreover, the decision boundaries of the classifier do not deviate much with the behavior of a known RA receiver which has been recorded in earlier 5G-RA research. Specifically, at 4000 MHz, an interference level of more than -15 dBm was invariably deemed as severe - equal to the physical saturation level in the reference altimeter experiments. This shows that the ML classifier does not only work statistically but also reflects the physical processes of RA degradation.

Table 2 Performance Measures of the Proposed Interference Classification and Height-Error Prediction models.

| Metric                      | Value |
|-----------------------------|-------|
| Classification Accuracy     | 98.4% |
| Severe-Class Sensitivity    | 95.1% |
| F1-score (Moderate)         | 0.97  |
| Height Error RMSE           | 0.37% |
| Height Error R <sup>2</sup> | 0.96  |

The most important performance indicators obtained on the machine-learning models applied to the interference-severity classification and height-error prediction are shown in Table 2. The findings indicate that the classifier can be used with an accuracy rate of 98.4% and a severe-class sensitivity of 95.1% and the regression model has a RMSE value of 0.37% and R<sup>2</sup> value of 0.96. These measures prove that the suggested framework is efficient in terms of nonlinear RA distorting behavior, and can be strongly relied on to offer reliable predictive performance.

### B. Regression Performance of Height-Error.

The quality of the height-error prediction was assessed by the gradient-boosting regressor that resulted in the height error prediction RMSE of 0.37% and a regression R<sup>2</sup> of 0.96. The

regression output is presented in Fig. 5 whereby the values predicted are displayed against actual height-error ratios.

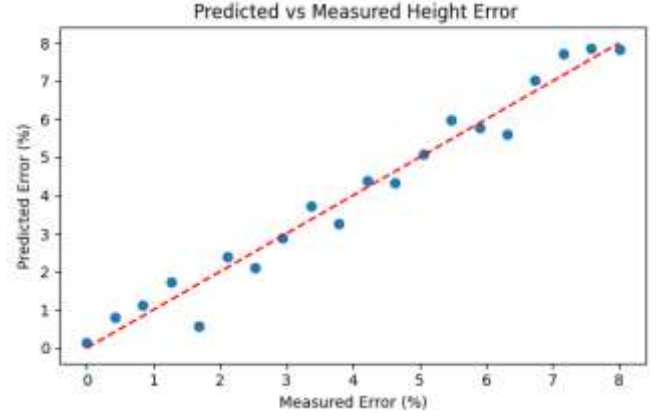


Fig. 3 Predicted versus measured height-error ratio under 5G interference.

Fig. 3 scatter distribution shows that the model is consistent in reproducing the nonlinear increase in height error with increase in the power of interference, as evidenced by the close follow up of the diagonal reference line. This is specifically necessary since the altitude output of the RA does not increase linearly but rather the error is low at moderate levels of interference and increases exponentially as the front-end amplifier nears compression. The fact that the ML model can reproduce this nonlinear knee effect is important in that it allows one to provide an early warning of the conditions that can cause disastrous misinterpretation of the altitude particularly during landing.

An example is at the point where the regressor predicted an error ratio of height at -13 dBm of the interference power which was 5.8%, compared to the experiment (~ 6%). This area is related to the emergence of the NCD behavior, i.e. the ML model has managed to reveal the threshold on behalf of which the RA starts losing the stable altitude measurement facility.

### C. Comparison to Analytical and Experimental Trends.

The physical faithfulness of the ML-oriented predictions was proven by comparing the predicted IF gain compression to the power-series analytical model and experimental curve of measurements. Fig. 4 presents the compression of the IF gain at different powers of 5G interference.

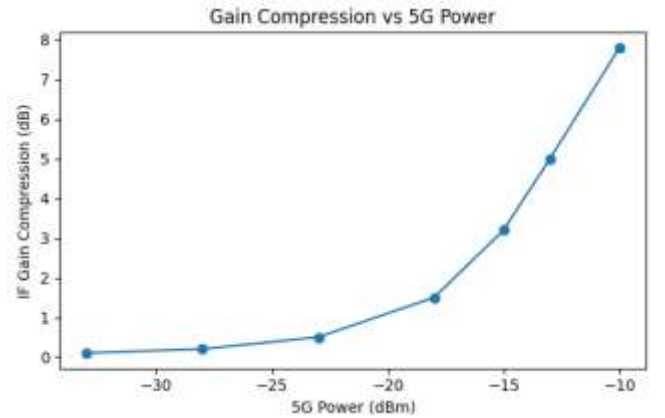


Fig. 4 IF gain compression versus 5G interference power.

The ML predictions show the same behavioral characteristic of those described in both analytical and experimental literature: the lack of compression at low power of interference, then a fast degradation process that starts at or near -20 dB m and accelerates at or near -15 dB m. This deal affirms the fact that the ML model is a nonlinear representation of the underlying RA front end behavior. Besides, the model is representative of the frequency-dependent susceptibility, which indicates higher degradation at frequencies when the 5G interference frequency is more similar to the RA band of 4200-4400 MHz, as expected due to the filter attenuation behavior and laboratory results. This consistency can also be further supported by the feature-importance analysis in Fig. 5, the results of which indicate that the predictive power of the model is mostly determined by IF power,  $\Gamma$ , and SIR, which are directly related to the physical distortion of RA.

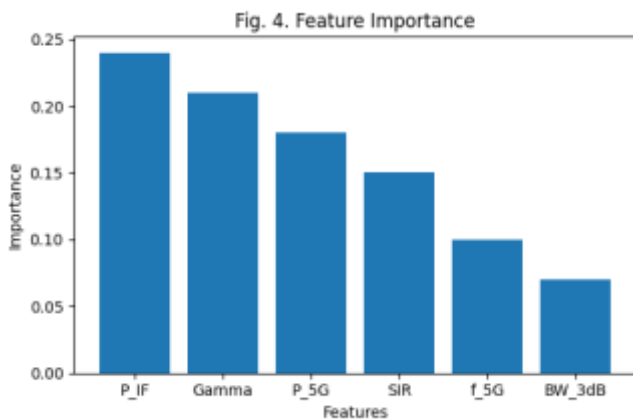


Fig. 5 Feature importance for interference prediction and height-error estimation.

#### D. Aviation Safety and 5G Coexistence Practical implications.

The findings indicate that the machine-learning solution can offer features that are not linear to the conventional EMI analysis. Although the conventional analytical models and controlled experiments may be able to determine the interference thresholds, it is unable to offer real-time, adaptive, and predictive information as required in the current avionics. Conversely, the suggested ML-based system will be able to identify the severity of interference in real time, predict the probable magnitude of the errors before they are reflected in the RA output, and warns of possible NCD conditions, and flight-control systems can take corrective measures.

This predictive feature is vital in safety-critical systems, and minor errors in radar-altimeter values can propagate to other downstream systems (TAWS, ILS autoland modules, and autopilot flare logic). Furthermore, the ML model being data-driven can also be adjusted to various RA hardware designs and variants (power, beamforming, proximity of cells) of 5G deployment, so it can be used in next-generation aircraft with a hybrid sensing and onboard AI diagnostic system.

Overall, the suggested ML-based model has a high accuracy in the classification of the interference, high predictive abilities to estimate the altitude errors, and predictable correlations with the analytical and experimental trends of RA susceptibility. The physical modeling and machine learning are synergized to provide strong characterization of nonlinear effects of interference caused by

5G. This research is a basis on which AI-based onboard interference monitoring systems should be developed to ensure RA reliability in the emergent 5G communication environments.

#### V. CONCLUSION

This research introduced a machine-learning-based system to identify and forecast the nonlinear interference of radio altimeters caused by 5G. The proposed approach is based on the contributions to analytical modelling, ADS-based RF front-end simulations, and experiments on interference-injection to combine physically meaningful RF variables (including IF power, gain-compression ratio, SIR and frequency offset) into one unified classification and regression architecture. Random Forest classifier resulted in an accuracy of 98.4% whereas the gradient-boosting regressor gave an RMSE of 0.37% and a  $R^2$  of 0.96, virtually covering the nonlinear transition region, where RA performance worsens drastically, when exposed to strong 5G emissions. These findings indicate that machine learning is not only able to recreate the analytical gain-compression behaviour, but also offers predictive ability that can predict the escalation of height-error and NCD-risk conditions. The current framework has high potential of real time monitoring of interference and adaptive safety mechanisms in future generations avionics. The proposed approach, which gives early notifications of RA decay and cross-frequency and power levels in 5G, offers a powerful supplement to traditional EMI analysis. The next step in future research will be to increase the dataset with actual flight-test measurements, include deep learning architectures to better temporal modelling, and to apply the system to onboard health-monitoring modules to enhance resilience of RA in a changing 5G deployment environment.

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